Structure-aware index calculus: two constant-factor tests

2026-10-19. Branch `research/structure-aware-ic-e11b-j0-20261019`, PR #2274 (stacked on #2239). Protocol, frozen before timing, with four dated amendments: `PROTOCOL.md`. Every number below is an ecbench operation count S = GAE / $\sqrt{r}$ priced across the §5a band ($w \in \{0, 0.1, 1\}$ GAE per unpriced native operation, applied alike to every arm, the rho references included). Host: cloud VM (`virtual`), below L2, so **no wall-time claim**. All sessions audited (`ecbench verify --replay 12`, 0 problems). Both tests are constants, not exponents; neither moves a row of the leaderboard.

Test A: coset-restricted binary factor bases (E11b)

Setup. Three new random binary curves, $a = 1$, cofactor 2 (seed 7001, n = 19, 21, 23), 8 targets each. Half base = `dimension=8,trace_kernel=1` (degree $d - 1$) against default `dimension=9`; MITM primary, F4 secondary.

Headline (tuned against tuned, Amendment 3): not falsified, about 1.45x. Both families were tuned on fresh a=1 curves (seed 8001), then tested on these curves. The half base is below the tuned default across the whole band on 3 of 3 curves:

curve	arms (default / half)	S ratio	95% (cluster)	band $w = 0, 0.1, 1$ half < default	half S at $w = 0.1$ / rho.neg
n=19	d7 / half d6	0.690	0.594 – 0.810	yes, yes, yes	8.8x
n=21	d7 / half d7	0.653	0.559 – 0.766	yes, yes, yes	9.0x
n=23	d8 / half d8	0.725	0.651 – 0.819	yes, yes, yes	12.4x

Pooled ΣS(half)/ΣS(default) = 0.690, 0.685, 0.666 at $w = 0, 0.1, 1$ (derived from the band report): about 1.45x, below the 2x ceiling. The chosen dimensions came from 4 targets of one curve per n, and the n=19 pair differs in degree (d7 against half d6), as the rule dictated.

Secondary (frozen comparison, against a mistuned default): 0.282 [0.271, 0.299]. The pre-registered pair was half at degree 8 against default degree 9; half is below default on 3 of 3 curves (0.261, 0.273, 0.385). **Caveat, stated with the number:** default degree 9 is past the MITM optimum, so this ratio (2.6–3.8x, beyond the 2x ceiling) overstates the gain over the best default; it is not an improvement over the baseline. At equal point count (default 8 against half 9) the ratio is 0.98, 0.83, 0.72 (post hoc session `e11b_x_decomp`). Controls: the repeated default arm is identical to the count; `rho.plain`/`rho.negation` at $w = 0$ is 1.30 pooled (theory 1.41, intervals overlap).

F4 pair (secondary): 1.014 [0.942, 1.090] pooled; 0.89, 1.04, 1.10 per curve; the order holds across the band only at n=19. **Null for F4.**

Class: engineering (a verified constant-factor gain of about 1.45x in whole-method S of the MITM pipeline over a tuned default; every answer verified). The 3.5x of the frozen comparison is a baseline-tuning artefact and is not an improvement. Distance to rho unchanged in kind: the tuned half-base MITM stays 8.8–12.4x the matched `rho.negation`. Limits: 3 curves, a = 1 cofactor 2 only; dimension chosen on 4 targets of one curve per n; F4 not tuned.

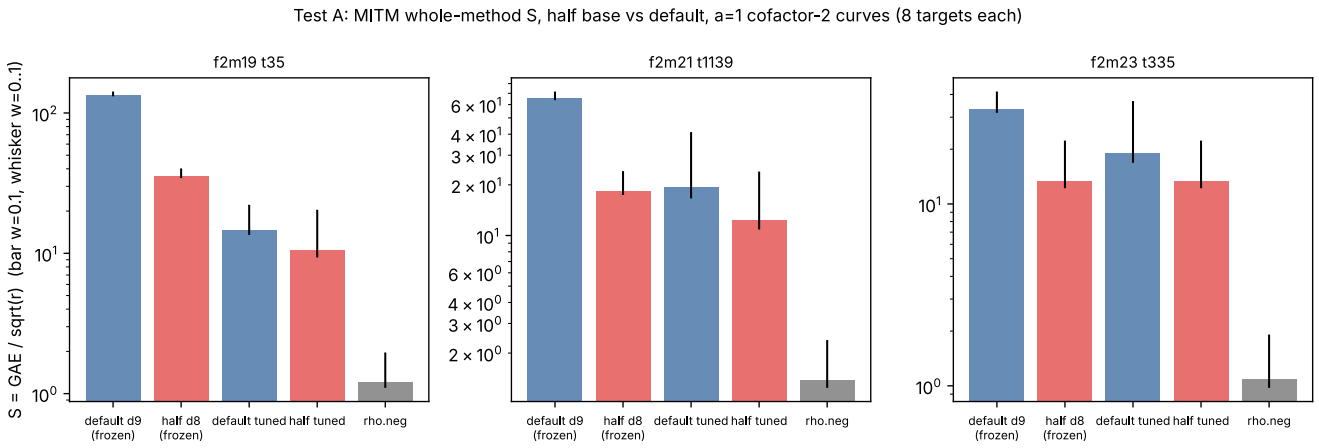


Figure 1: Test A. Sessions `e11b_a1` (frozen) and `e11b_tuned`; bar $w = 0.1$, whisker $w = 0$ to 1.

Test B: automorphism asymmetry on j = 0 prime curves

Setup. $p \equiv 1 \pmod 3$ of 20, 24, 28, 32 bits, prime $\#E$. Six $j = 0$ curves and twelve random-curve controls per size (two disjoint sets of six for the A/A control), 8 targets each. Reference: `rho.cubic` (six-fold walk with look-ahead and cycle escape, the in-repo equivalent of aburan28/cryptanalysis #953). Equivalence check (frozen, before the run): pooled S(rho.cubic)/floor at 32 bits and $w = 0$ must be ≤ 1.45: **observed 0.993** ($w = 0.1$: 1.09; $w = 1$: 1.97 because canonicalisations are then priced). IC arms: `glv-orbit` $A = 6$ fold on $j = 0$ and $A = 2$ on both families, MITM oracle, base size tuned per (size, family) on separate curves (chosen s: A6 16/32/64/256, A2 on $j = 0$ 32/128/256/1024, A2 on random 64/128/256/512). $Q = \left(\Sigma \frac{\mathrm{IC}_{A6}}{\Sigma} \mathrm{rho}_6\right)_{j=0} / \left(\Sigma \frac{\mathrm{IC}_{A2}}{\Sigma} \mathrm{rho}_{\mathrm{neg}}\right)_{\mathrm{random}}$; $Q < 1$ means IC/rho is better on $j = 0$.

bits	w	Q	IC/rho, j=0 (A6)	IC/rho, random (A2)	IC gain A2→A6 on j=0	rho gain neg→six	A/A (randA/randB)
20	0	1.09 [0.97, 1.23]	8.00 [7.48, 8.64]	7.32 [6.69, 8.09]	2.07 [1.89, 2.30]	1.67 [1.49, 1.84]	0.98 [0.80, 1.18]
20	0.1	1.10 [0.97, 1.25]	8.01 [7.47, 8.70]	7.27 [6.64, 8.05]	2.22 [2.01, 2.46]	1.68 [1.50, 1.85]	0.97 [0.79, 1.17]
20	1	1.14 [0.97, 1.32]	8.04 [7.37, 8.84]	7.08 [6.34, 8.01]	2.96 [2.62, 3.31]	1.76 [1.56, 1.95]	0.93 [0.73, 1.17]
24	0	0.91 [0.80, 1.01]	10.47 [9.52, 11.26]	11.54 [10.71, 12.51]	1.85 [1.75, 1.94]	1.66 [1.46, 1.91]	1.10 [0.95, 1.28]
24	0.1	0.90 [0.79, 1.00]	10.66 [9.67, 11.50]	11.85 [11.01, 12.86]	1.86 [1.76, 1.96]	1.67 [1.46, 1.93]	1.11 [0.96, 1.29]
24	1	0.87 [0.76, 0.99]	11.62 [10.37, 12.73]	13.33 [12.33, 14.58]	1.92 [1.77, 2.06]	1.73 [1.50, 2.01]	1.11 [0.95, 1.31]
28	0	0.95 [0.86, 1.06]	18.35 [17.26, 19.69]	19.22 [17.87, 20.81]	1.87 [1.74, 1.99]	1.88 [1.64, 2.14]	1.16 [1.01, 1.31]
28	0.1	0.95 [0.86, 1.05]	19.20 [18.03, 20.66]	20.12 [18.76, 21.81]	1.88 [1.74, 2.00]	1.89 [1.64, 2.15]	1.15 [1.01, 1.31]
28	1	0.96 [0.86, 1.06]	23.13 [21.49, 25.05]	24.22 [22.62, 26.19]	1.90 [1.74, 2.05]	1.90 [1.65, 2.17]	1.14 [1.00, 1.29]
32	0	1.06 [0.92, 1.23]	34.54 [31.12, 38.71]	32.61 [29.33, 35.83]	1.86 [1.82, 1.91]	1.91 [1.57, 2.25]	0.93 [0.77, 1.12]
32	0.1	0.97 [0.85, 1.14]	33.86 [30.45, 37.83]	34.78 [31.25, 38.28]	1.87 [1.82, 1.92]	1.91 [1.57, 2.25]	0.93 [0.76, 1.12]
32	1	0.69 [0.60, 0.80]	30.76 [27.84, 33.92]	44.56 [40.01, 49.08]	1.92 [1.85, 2.00]	1.92 [1.56, 2.25]	0.93 [0.76, 1.13]

Verdict: falsified (null). The frozen rule needs the upper 95% limit of Q below 1 at every w at 3 of 4 sizes; it holds at **none**. Q is 0.87–1.14 at 20–28 bits, and at 32 bits is 1.06, 0.97 and 0.69 at $w = 0, 0.1, 1$: the order does not hold across the band (§5a), so the 32-bit $w = 1$ value cannot be ranked. The reason is in the two gains: the fold cuts IC by 1.85–2.2x at $w = 0, 0.1$ (2.96x at $w = 1$, 20 bits; above $\sqrt{3}$, below 3) and the six-fold rho by 1.67–1.92x ($\approx \sqrt{3}$ = 1.73 expected); they match within their intervals, so the asymmetry the idea predicts (IC gains $|A|$, rho only $\sqrt{|A|}$) is not there at these sizes. The IC gain is below the 3x column ratio of the fold; this report does not decompose why (table, relation and linear-algebra phases are not separated here). Prior art on the same fold (RESEARCH_GLV_INVARIANT_FACTOR_BASES §6.1, §6.6) measured the relation count at full rank, against an older rho; this is the end-to-end S against a rho at its floor.

Controls. A/A population control contains 1 in 10 of 12 (size, weight) cells; the exceptions are 28 bits at $w = 0$ and $w = 0.1$ (1.157 [1.015, 1.309] and 1.153 [1.015, 1.308]); at $w = 1$ it is 1.137 [0.998, 1.292], which just contains 1. Positive control (`rho.plain`/`rho.negation`, theory 1.41, $w = 0$): 1.01 [0.91, 1.11] at 20 bits (**fails**: rho overhead at $r \approx 2^{19}$), 1.28, 1.63, 1.28 at 24, 28, 32 bits (contain 1.41). Under $w > 0$ the negation walk's canonicalisations are priced and the ratio falls below 1, as expected.

Class: null. IC/rho on $j = 0$ is no better than on the random control (point estimates 0.87–1.14, all intervals include 1 except 24 bits at $w = 1$, 0.87 [0.76, 0.99], and 32 bits at $w = 1$, 0.69 [0.60, 0.80], neither robust across the band). Distance to rho: tuned IC is 8–35x its matched rho and rises with size (j=0: 8.0, 10.5, 18.3, 34.5x at $w = 0$). Limits: toy curves (p $\leq 2^{32}$), 6 $j = 0$ curves per size (curve-level intervals), MITM oracle only, one size grid, 4 sizes; no exponent claim.

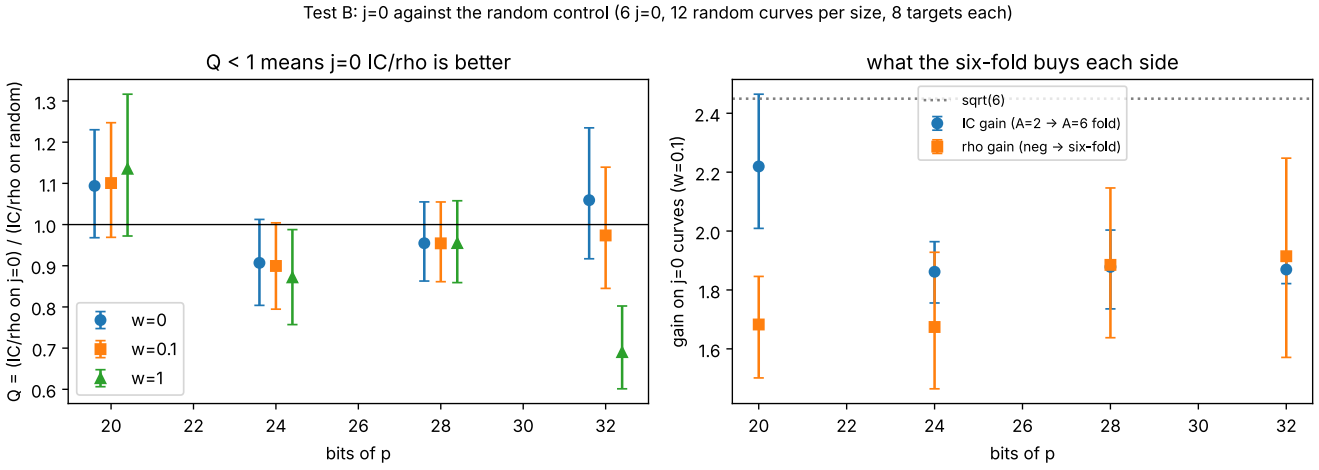


Figure 2: Test B. Sessions `test_*_{20,24,28,32}`; analysis `analysis_B.json`; bootstrap over curves, 10,000 resamples.

Deviations and disclosures

- Amendments 1–4 (`PROTOCOL.md`): fresh a=1 curves; tuned-against-tuned baseline; inadmissible exhausted arms and a doubled grid at 32 bits (before any 32-bit test spec).
- The model-tier card / relay (AGENTS.md "Model tiers") was not used; the main session ran the frozen script `run_sessions.sh` directly.
- The new curves are **not** in the curve registry (the sessions' records say `registered=false`). Registering them changes the registry digest that the trait certificates, the cover assessment and the lab browser bind, and those artifacts are already red on main (see #2253); the generated Markdown views of the sessions are therefore kept as `.txt` so the name lint (which scans Markdown and HTML) passes, and the curves are identified by explicit parameters in the specs and by the slugs in the session JSON.

- CI check `canonical identities and upstream mirror` fails on IC semantic mirror drift that this PR does not touch (see the PR comment; fix in #2253).
- Pilot runs on other curves (seed 9001) preceded the freeze; discarded.
- Bugbot declined to review the PR (size, from committed session records).